Agentic AI in mathematical research

A brief disclaimer

I am not an expert in this field. I have been exploring it out of curiosity and to stay up to date.

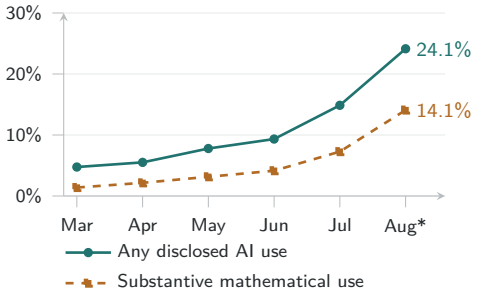
This talk neither advocates nor argues against using AI to write papers.

- A brief history and overview of mathematical research with LLMs, focusing on agentic approaches.
- What the literature tells us about successful approaches and their limitations.
- A case study where I build an AI research team to write a paper.

AI use and mathematical output

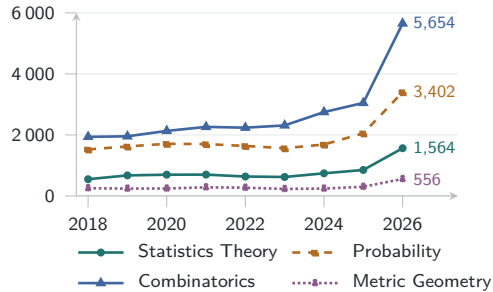
AI disclosures on arXiv

Share of math-related submissions, 2026



Submissions by field

New preprints, January–August each year

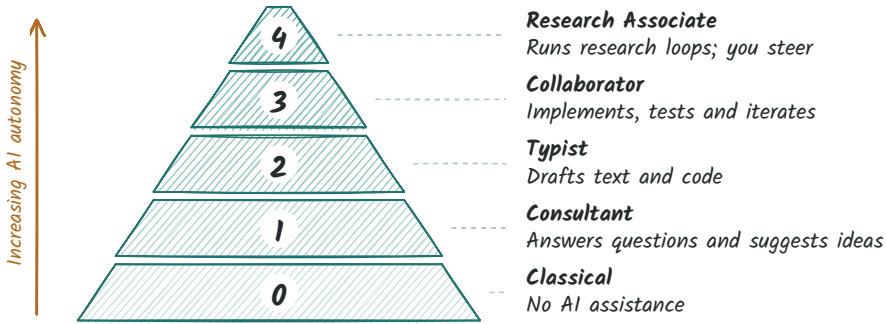


Left: 3,575 of 32,944 preprints disclosed AI use. *August covers 1–20.
Right: primary-category counts. Growth alone does not identify an AI effect.

Jin, Ke & Sui (2026), Table 1 **Explore arXiv Math Pulse** GitHub source.

How have you used AI in research?

Which is the highest level you have used?



Adapted from Zimmer et al. (2026), *The Agentic Researcher*, Section 2.

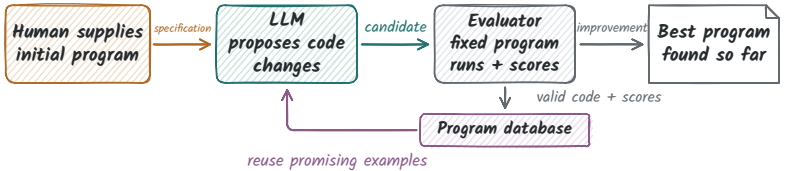
A short history of math research with LLMs

Agentic research systems over time

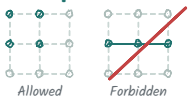
A brief history, with a focus on mathematics.

FunSearch Romera-Paredes et al., *Nature* (2024)

An evolutionary approach with fixed evaluation criteria implemented in computer programs.



Larger cap sets



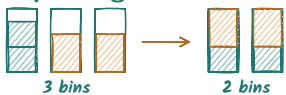
496 → 512

points in 8 dimensions

No three distinct points on a line in $\mathbb{F}_3^n$.

A new construction, not a proof of maximality.

Online bin-packing heuristics



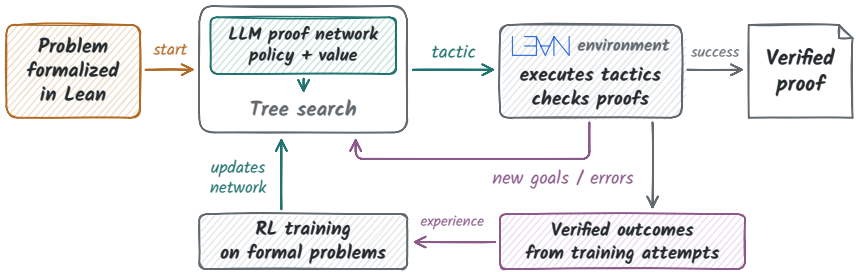
Use as few fixed-capacity bins as possible; place each item on arrival.

AlphaEvolve (2025) later extended this approach: 4×4 complex matrix multiplication with **48 scalar multiplications**, versus Strassen's 49.

Workflow: Fig. 1; results: Fig. 4 and Table 1. Romera-Paredes et al. (2024).

AlphaProof Hubert et al., *Nature* (2026)

Another way to verify and evaluate mathematics is through formal verification in Lean.



Search: the LLM proposes tactics; Lean returns new proof states.

Learning: The model learns from proof attempts. For harder problems, it practises on similar ones.

IMO 2024: silver-medal level — 28/42 points with AlphaGeometry 2.

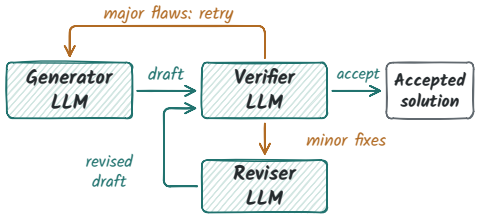
AlphaProof solved 3 problems; AlphaGeometry 2 solved 1.

Adapted from Figs. 1–2 and the AlphaProof section. *Nature* **651**, 607–613 (2026).

Aletheia Feng et al., arXiv (2026)

We can also use an independent LLM for verification. This does not guarantee correctness.

Gemini Deep Think agents



LLM judgment; no formal guarantee (Fig. 1).

Reported mathematical results

- Erdős: 2 full + 2 partial new solutions from 700 problems, assessed by experts.

And other mathematical achievements.

Other LLM research systems

The AI Scientist *Nature* (2026)
Ran ML experiments and wrote a paper that passed a workshop's first review round.

Google Co-Scientist *Nature* (2026)
Agents proposed and refined leukaemia drug combinations, validated in laboratory experiments.

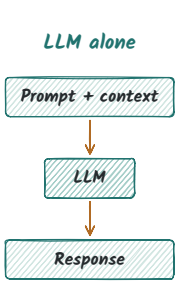
Virtual Lab *Nature* (2025)
LLM agents designed SARS-CoV-2 nanobodies with binding confirmed in laboratory tests.

OpenAI: Navier–Stokes report (2026)
LLM teams explored different approaches; the successful team ran about **10,000 agents** concurrently. **Human researchers** redirected effort and used Codex to share insights between teams.

Agentic systems

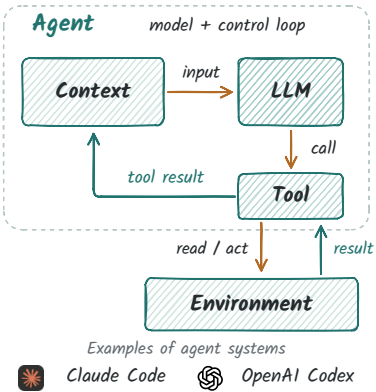
From LLMs to research teams

Environment, context, tools and agents



$$Y = f_\theta(p, c; \xi)$$

p: prompt; c: context
θ: fixed model
ξ: randomness



Environment
The workspace an agent can access: files, data, code and external services.

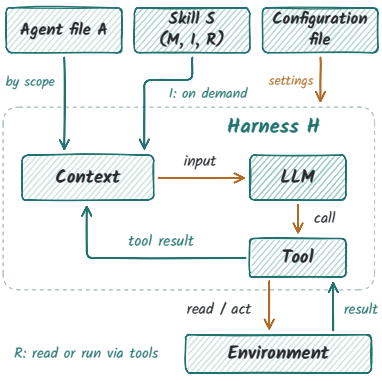
Context
The information given to the LLM now:
instructions, messages and tool results.

Tool
An interface to observe or change the environment, e.g. search or Python.

Agent
An LLM-based system that uses tools and chooses its next step from results.

Based on OpenAI's Codex harness documentation and Claude Code's agentic loop.

Skills, agent files and the harness



Agent instruction file $A = (D, I_A)$
D: scope; I_A: standing instructions.
E.g. `AGENTS.md`: project guidance loaded into the context within its scope.

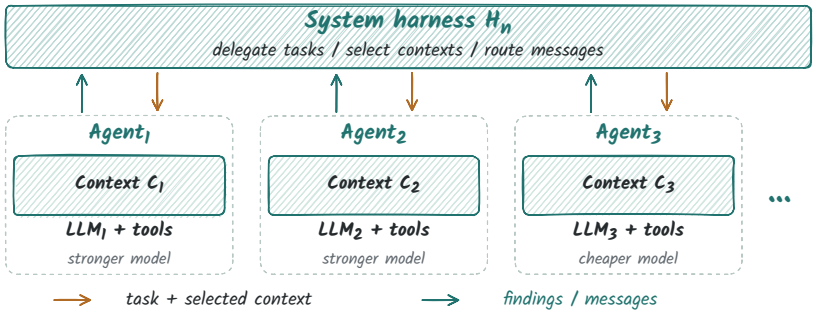
Skill $S = (M, I, R)$
M — **metadata**: name, what the skill does, and when to use it.
I — **instructions**: method, decision rules, checks and expected output.
R — **resources**: optional documents, code, examples and templates.

Harness H
Orchestration software. A configuration file sets roles, models, tools, skills and routing.

Working abstractions for this talk. Implementation: skills, `AGENTS.md`, Codex harness (OpenAI Docs).
Foundations: ReAct — Yao et al. (2022), Voyager — Wang et al. (2023), SWE-agent — Yang et al. (2024).

Multi-agent systems: who sees what?

Recall $Y = f_\theta(p, c; \xi)$. So context engineering is very important.



H_n is the software coordinating n agents, their contexts and communication.

A configuration file sets roles/models/tools; skills and agent files supply instructions.

Same or different models. Sources: AutoGen (Wu et al., 2023); OpenAI agent configuration.

Why manage context and communication?

Very simply, an LLM produces random text $Y = f_\theta(p, c; \xi)$, whose distribution **depends strongly on the context**.

1. Keep context focused Liu et al. (2024); Modarressi et al. (2025)
Problems: context distraction and context clash.
Give each agent the information it needs; long histories can obscure relevant details.

2. Give review a fresh context Sharma et al. (2023); Dhuliawala et al. (2023)
Problems: context poisoning and context confusion.
Review the draft and criteria without the writer's history; verify independently.

3. Match the model to the task Ong et al., *RouteLLM* (2024)
Routing between stronger and cheaper models can improve the balance of cost and quality.

4. Share findings and avoid duplicate work Du et al. (2023); Wang et al. (2025)
Give each agent the results it needs. Redirect repeated work when it adds no new information.

Effects depend on the task and models. Fresh-context review is a design inference from bias and verification studies.

Heuristics for agentic research

Heuristic	Intuition
Explore distinct approaches independently. [1]	Different starting points can reveal approaches that others miss.
Fix identified errors; stop if no progress is made. [2]	Check that each error was repaired. Repeated approval is not evidence of a correct proof.
Ask for a proof or a counterexample. [3]	The statement you give the agent might be false. Explicitly allow it to disprove the statement instead of assuming a proof must exist.
Delegate precise mathematical obligations. [4]	When splitting the work between agents, specify exactly what each agent needs to establish and under which assumptions.
Preserve each claim's status and evidence. [5]	Reuse must not turn a conjecture or rejected argument into an "established lemma".

Evidence-informed design heuristics: support ranges from benchmark experiments to research-system case studies.

[1] Zhu et al. (2026); [2] Zheng et al. (2026); [3] Petrov et al. (2025);
[4] Zhao et al. (2026); [5] Xu et al. (2026).

Agentic research: a worked example

What do we need?

- An **agent system**: in my case, Codex.
- A precise **mathematical problem statement**.
- Relevant **related work**: definitions, known results and useful methods.
- **`instruction.md`**: the task, constraints and expected output.
- **`skills.md`**: an index of reusable methods and checks.
- **`AGENTS.md`**: standing project instructions for the agents.

Here, `instruction.md` and `skills.md` are our project documents.
Each actual Codex skill has its own `SKILL.md`.

Point processes: a Poisson example

A **simple point process** X is a random set of distinct points in $\mathbb{R}^d$, finite in bounded regions.

Its count in a region B is $X(B) = \#(X \cap B)$.

Stationarity

Shifting all points leaves the distribution unchanged:

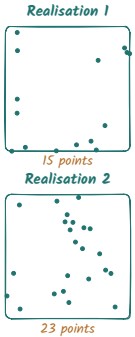
$$X + t \overset{d}{=} X \quad \text{for every } t \in \mathbb{R}^d.$$

Homogeneous Poisson process, intensity $\lambda > 0$

$$X(B) \sim \text{Poisson}(\lambda|B|), \qquad \mathbb{E}[X(B)] = \lambda|B|.$$

Here $|B|$ is the volume of B; λ is the mean number of points per unit volume.

Counts in disjoint regions are **independent**.

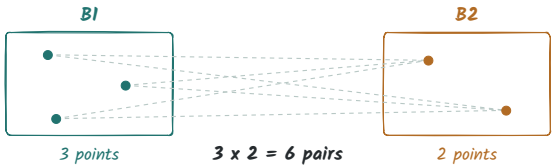


Unit-area windows; $\lambda = 20$.
On average 20 points per window.

Baddeley, Sections 1.4–1.6. Pictures show independent simulated realisations in two dimensions.

Product densities: counting pairs

Take two **disjoint** regions B_1, B_2. The product $X(B_1)X(B_2)$ counts pairs with one point in each region.



Across repeated realisations, for small regions around x_1, x_2:

$$\underbrace{\mathbb{E}[X(B_1)X(B_2)]}_{\text{expected number of pairs}} \approx \underbrace{\rho^{(2)}(x_1, x_2)}_{\text{product density}} |B_1||B_2|.$$

Product density: local abundance of pairs, after accounting for the sizes of the two regions.

Three or four regions: count **triples** or **quadruples** in the same way.

The approximation applies where the density varies little across the small regions.
Baddeley, Section 2.3; Last and Penrose, Section 4.4.

Product densities: comparison with Poisson

For a homogeneous **Poisson process**, independence gives

$$\rho^{(2)}(x_1, x_2) = \lambda^2, \qquad \rho^{(n)}(x_1, \ldots, x_n) = \lambda^n.$$

For a **stationary process** with intensity $\lambda > 0$, compare its product density with this reference:

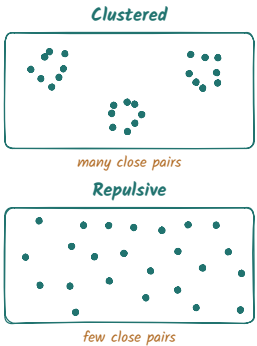
$$g_n(x_1, \ldots, x_n) = \frac{\rho^{(n)}(x_1, \ldots, x_n)}{\lambda^n}.$$

This is the *n*-**point correlation function**.

Compared with Poisson at the same intensity:

$g_n > 1$: **more such tuples.**
$g_n < 1$: **fewer such tuples.**
$g_n = 1$: the same local tuple density.

$n = 2$: pairs; $n = 3$: triangles; $n = 4$: tetrahedra.

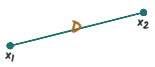


Schematic point patterns.

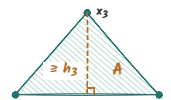
Comparisons concern the specified locations and scale. $g_2 = 1$ alone does not imply Poisson.
Baddeley, Section 2.3; Last and Penrose, Section 4.4.

Step 1: the precise problem

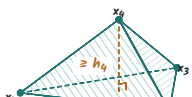
Fix $M, \lambda_0, h_3, h_4 > 0$.
Find **one stationary, simple** point process X on $\mathbb{R}^3$ with intensity λ_0, finite local moments through order four and product densities.



Distance D



Area A



Volume V

For some constants $C_2, C_3, C_4 > 0$, require **simultaneously**:

$$g_2(x_1, x_2) = \frac{C_2}{D}, \qquad g_3(x_1, x_2, x_3) = \frac{C_3}{A}, \qquad g_4(x_1, x_2, x_3, x_4) = \frac{C_4}{V}.$$

Almost everywhere on nondegenerate configurations of diameter $\leq M$, with every triangle altitude $\geq h_3$ and every tetrahedron altitude $\geq h_4$.

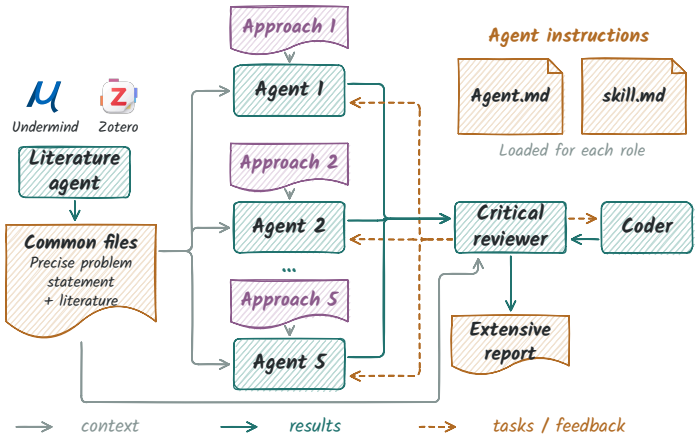
C_2, C_3, C_4 are chosen jointly with the process, not prescribed independently.

Current construction: sufficiently small λ_0; arbitrary prescribed intensity remains open.

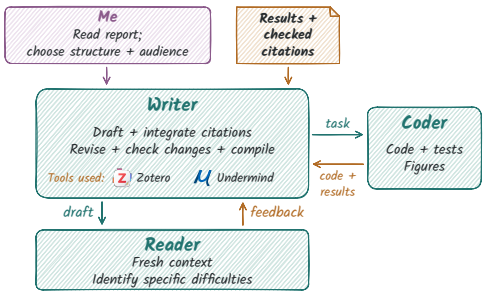
Step 2: configuration/agent files and execution

Research approaches

1. Poisson cluster process
2. Cox process
3. Hawkes process
4. An approach of the agent's choice
5. Prove non-existence



Step 3: Writing the paper



Complete-task statistics

Measure	Complete task
Input tokens	131,866,681
Cached input (included above)	128,329,472
Uncached input	3,537,209
Output (includes reasoning)	666,705
Reasoning (subset of output)	286,459
Total tokens	**132,533,386**
Task-turn runtime	**3:47:06**
Research coordinator runtime	**3:36:43**
API-equivalent cost	**$78.81**

Writing stage: token use and cost

Participant	Uncached input	Cached input	Output/reasoning	Cost
Coordinating writer	0.30M	3.5M	0.15M	$5.60
Simulation specialist	0.15M	1.4M	0.10M	$3.16
Fresh reader	0.06M	0.1M	0.02M	$0.68
Total	**0.51M**	**5.0M**	**0.27M**	**$9.44**

Results and personal feedback

Poisson cluster model

Homogeneous Poisson parents Π each generate **0–3 children**, with jointly chosen offsets C_z:

$$X = \bigcup_{z \in \Pi} \big(\{z\} \cup (z + C_z) \big).$$

C_z contains only child offsets. **The parent is retained**, giving 1–4 points per cluster.

How many points? The solution gives a probability distribution over cluster sizes $j = 1, \ldots, 4$.

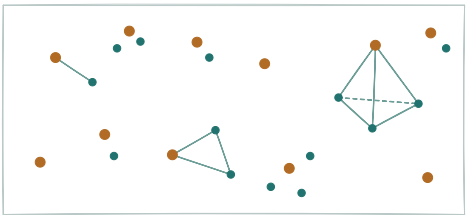
Where are they placed? For $j \geq 2$, the normalised density

$$\frac{F_j(z, x_1, \ldots, x_{j-1})}{Z_j}$$

describes how the $j - 1$ children are jointly distributed around the parent z.

Personal feedback

Aspect	Feedback
Mathematical correctness	Positive
Research and process documentation	Positive
Code writing	Positive
Text writing	Medium
Paper structuring	Negative
Mathematical explanation	Medium



Future of agentic research

What I expect to change

Stronger models can increasingly plan, divide work into tasks, use tools and delegate to subagents.

For a precise mathematical problem, **one initial prompt may increasingly suffice**. This can still mean substantial autonomous work, with no guarantee of success.

As platforms provide more harness functions, we will have **less to build ourselves**.

I also expect explanations to become **less sloppy and more useful for learning**.

Where a custom harness still helps

- Keep the exact target, hypotheses and criteria for success fixed.
- Require separate review and formal verification.
- Keep human checkpoints for direction and interpretation.
- Record results, failed approaches, sources and dependencies.
- Set compute budgets, stopping rules and model choice.
- When writing papers with AI, set your preferences for writing and illustration styles.

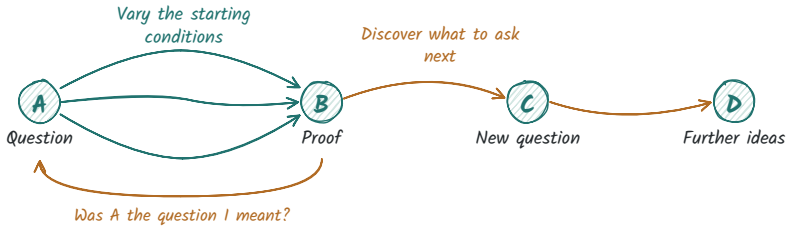
A reported example. For its October release, OpenAI says almost all results came from one prompt to one agent. Some may have needed retries; the full workflow has not been independently reproduced.

Scientific American, October 2026; OpenAI release details.

Personal take

I find this exciting: an elegant solution, with reasoning that feels human.
Still, something is missing when I do not discover the ideas myself.

Faster proofs let me explore more. Research was never a straight line.



Reaching B only helps me if I understand the mathematics.

My concern as a PhD student: outsourcing too much may bypass the learning.
A PhD is still an education. Producing results and developing understanding can come apart.

Related: *A Severe Misalignment of AI in Mathematics* (joint declaration, 11 September 2026).

Thank you!

Questions and discussion

Slides and, later, the experiments:

github.com/MathisRost/agentic-ai-math

